Unit 3 Pre-Test

Solving Linear Equations

Standard assessed: NC.8.EE.7

Name: _____ **Date:** _____ **Period:** _____

SCORE **3.0** **Basic Understanding**

NC.8.EE.7 Solve linear equations in one variable.

- *I can solve one-step and two-step equations using inverse operations.*

Match each vocabulary term to its definition, then solve each equation.

C coefficient ✓	**A.** Terms with the same variable raised to the same power.
E constant ✓	**B.** The value of the variable that makes the equation true.
D inverse operations ✓	**C.** A number multiplied by a variable.
B solution ✓	**D.** Operations that undo each other, such as + and −.
F distributive property ✓	**E.** A term with no variable — just a number.
A like terms ✓	**F.** Multiplying a factor across each term inside parentheses.

Keep the equation balanced — worked example

$2x + 5 = 13$ → *subtract 5* → $2x = 8$ → *divide by 2* → **$x = 4$**

Whatever you do to one side, do to the other.

1. *Solve*

$x + 7 = 12$

2. *Solve*

$x - 4 = 9$

x = _____

answer: x = 5

x = _____

answer: x = 13

3. *Solve*

$3x = 21$

x = _____

answer: x = 7

4. *Solve*

$x \div 5 = 6$

x = _____

answer: x = 30

5. *Solve*

$2x + 3 = 11$

x = _____

answer: x = 4

6. *Solve*

$4x + 5 = 25$

x = _____

answer: x = 5

SCORE 4.0 Target Content

NC.8.EE.7 Solve equations with the distributive property and variables on both sides.

• *I can solve multi-step equations, including variables on both sides.*

Solve each equation. Show your work.

7. *Solve*

$2(x + 3) = 14$

x = _____

answer: x = 4

8. *Solve*

$3(x - 2) = 9$

x = _____

answer: x = 5

9. *Solve*

$5x + 2 = 3x + 10$

x = _____

answer: x = 4

10. *Solve*

$7x - 4 = 2x + 11$

x = _____

answer: x = 3

11. *Solve*

$2(x + 4) = 3x + 5$

x = _____

answer: x = 3

12. *Solve*

$3(2x - 1) = 4x + 7$

x = _____

answer: x = 5

SCORE 5.0 Advanced Content

NC.8.EE.7 Solve equations with fractions or decimals and identify special solutions.

• *I can solve equations with rational coefficients and recognize equations with one, none, or infinitely many solutions.*

Solve. If there is no solution or infinitely many, write that instead.

13. *Solve*

$\frac{1}{2}x + 3 = 9$

x = _____

answer: x = 12

14. *Solve*

$0.2x + 1 = 3$

x = _____

answer: x = 10

15. *Solve*

$\frac{2}{3}x = 8$

x = _____

answer: x = 12

16. *Solve*

$3x + 5 = 3x + 5$

Answer: _____

answer: infinitely many solutions

17. *Solve*

$2x + 4 = 2x - 1$

Answer: _____

answer: no solution

18. *Solve*

$5(x + 2) - 3 = 2(x + 5)$

x = _____

answer: x = 1